

Embedded systems Exp 16

Program:

```
#include <reg51.h>

#include <stdio.h>

void serial_ISR(void) interrupt 4
{
    char ch;

    if (RI)
    {
        RI = 0; // Clear receive interrupt flag

        ch = SBUF; // Read received character

        printf("\n>>> INTERRUPT RECEIVED: ");

        printf("%c\n", ch);
    }
}

void delay(void)
{
    unsigned int i;

    for(i = 0; i < 50000; i++); // Delay for visibility
}

void main(void)
{
    SCON = 0x50; // UART mode 1, 8-bit, REN enabled

    TMOD = 0x20; // Timer1 mode 2 (auto-reload)

    TH1 = 0xFD; // Baud rate 9600

    TR1 = 1; // Start Timer1

    IE = 0x90; // EA = 1, ES = 1 (enable serial interrupt)

    TI = 1; // Enable printf

    while(1)
    {
        printf("Hello World\n");
    }
}
```

```
delay();
```

```
}
```

```
}
```

Output

The screenshot shows the Keil uVision IDE interface. The main window displays the source code of a C program in `exp16.c`. The program includes a delay function and a `main` function that prints "Hello World" four times. The register window on the left shows the values of various registers, including `r0` through `r7` and system registers like `PC`, `SP`, and `PC`. The command window at the bottom shows the execution of the program, with the output "Hello World" printed four times.

```
19: void main(void)
20: {
21:   SCON = 0x50; // UART mode 1, 8-bit, REN enabled
22:   TH0 = 0x00; // Timer0 mode 1, 16-bit, auto-reload
23:   TR1 = 1; // Start Timer1
24:   IE = 0x90; // EA = 1, ES = 1 (enable serial interrupt)
25:   TI = 1; // Enable printf
26:   while(1)
27:   {
28:     printf("Hello World\n");
29:     delay();
30:   }
31: }
32: }
```

Command Window:

```
Running with Code Size Limit: 2K
Load "C:\Keil_v5\C51\Examples\HELLO\keil\Objects\exp16"
Hello World
Hello World
Hello World
Hello World
```